A family of quartics in 297 variables

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Abstract

We specify a family of integer quartics in 297 total variables by 271 quadratic equations and complete coefficient tables. Under the arithmetic premises of the elliptic source construction, every sufficiently large member defines the integers in the rationals. Two height detectors use the rational trace of a sum of elliptic abscissas. The sixteen local tests are represented by polynomial expressions with distinct valuation classes, using fourth and eighth powers already computed for the formal logarithm. The coefficient family is explicit, but the sufficient parameter threshold is ineffective because the aggregate-trace height theorem has an ineffective additive constant. No particular defining member of this family is selected. A fully specified 342-variable quartic is a separate result.

1 The exact family and its scope

All variable counts include the input X. Put

$$A_N = 80989^{2N+1} \qquad (N \geq 1).$$

The tables `quadratic-family297.json` and `polynomial-family297.json` specify 271 primitive integer quadratic rows $Q_{N,j}$ in 297 coordinates and

$$P_N = \sum_{j=1}^{271} Q_{N,j}^2.$$

Neither A_N nor N is a variable of P_N.

Theorem 1.1. *Under the arithmetic premises in Section 2, there is an integer $N_0 \geq 27464$ such that, for every $N \geq N_0$ and $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{296},\ P_N(x, \boldsymbol{y}) = 0.$$

For every $N \geq 1$, P_N has degree exactly four, content one, and coefficient height

$$H(P_N) < 2^{1520} 80989^{4N+2}.$$

For every $N \geq 1$ it has 4268 nonzero coordinate monomials. Its exact height is $80989^{4N+2} + 1$ for $N \geq 23$. The proof does not determine a numerical value of N_0.

An exact family specification and an effective choice of a defining member are different assertions. The theorem supplies the first together with eventual definition of $\mathbb{Z}$; its height argument does not supply the second. The explicit 342-variable theorem in [4] uses three height detectors and an effective sum-height inequality.

2 Arithmetic premises

Let $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, $d = 3 - \rho$, and

$$E : y^2 = x^3 - d^2x, \qquad P = 2(1 + \rho, 2\rho) = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32}\right).$$

The arithmetic input contract is the one precisely stated in [6, Section 2] and restated in [4, Section 2]:

1. $2E(K) = \mathbb{Z}P$ and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n.
2. The five-contact height bound $h(t) \leq \mathcal{H} M_5(t)^{19}$, with

$$\mathcal{H} = 2^{110} 210^{1800} \left(\prod_{p \leq 76176} p\right)^{19} + 128 < 2^{2093448}.$$

   Here $M_5(u/v)$ is the product of primes greater than 76176 with positive odd valuation on at least one of $v, u, u - v, u - 2401v, 32u + 343v$, in lowest terms with $v > 0$, and the excluded contact values are $0, 1, 2401, -343/32$.
3. The source pole obstruction outside $2, 3, 17$ and its simultaneous realization for inputs in $(0, 17)$ with square reduced denominator and the specified local squareclasses and valuation bounds at $2, 3, 17$. The normalized seven-coordinate equations are retained literally.
4. The simultaneous prime-pattern and conjugate reduction-order conditions with arbitrary fixed odd avoidance modulus, finite congruences, and real approximation, including the quantitative weighted selection premise stated in the cited contract. In particular the slowly decreasing separation parameter is available after all fixed data are chosen.

No finite coefficient computation proves these inputs. The weighted selection and separation used here follow from the cited analytic estimates as proved in [2]; no additional prime-distribution conjecture is introduced by the local weight replacement. The aggregate-trace height theorem used next is a proved consequence of Von Buhren's family height theorem, with an ineffective additive constant, as detailed in [7]. It is not an additional conjecture.

3 The trace height and the two local detectors

The retained square charts give root points

$$Q_1 = [n]P, \quad Q_2 = [a]P, \quad Q_3 = [t]P, \quad Q_4 = [b]P, \qquad B = \max(|n|, |a|, |t|, |b|),$$

with nonzero integer labels. Their interpretation for all nonzero integer labels, without a parity restriction, is proved in [5, Section 2]. Define

$$S_0 = \frac{1}{2} \operatorname{Tr}_{K/\mathbb{Q}} \left(\sum_{i=1}^{4} x(Q_i)\right).$$

There are fixed constants $M_1 > 0, M_2 \geq 1$ such that

$$B^2 \leq M_1 h(S_0) + M_2. \tag{1}$$

The additive constant is not effective in the cited proof. Every root abscissa is positive at both real embeddings, so $S_0 > 12$.

Let σ denote conjugation on $K/\mathbb{Q}$ and let $f = 1/x$, extended by zero at the identity. The auxiliary comparison points satisfy $U_0 = H$, $V_0 = H - P$, and $Q_i = U_i - V_i$ as group differences on E. These are the exact addition relations of the retained point charts in [5].

Use the two nonzero rational square scales $1, S_0^2$. For each scale a_* put $t_* = a_*\lambda$, $\delta = t_* - 1$ and

$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\},$$
$$R_* = \frac{\prod_{\theta\in\Theta}(t_* - \theta)}{a_*^5\delta^6}, \qquad q = R_*^{-1}\delta^{-4}, \qquad q_m = q^{2m-1}.$$

These formulas are used on their nonzero domain; the exact system consists of their cleared quadratic equations. The case $t_* = 1$ has empty contact radical and requires no reciprocal.

Define $A_i = f(U_i)$, $B_i = \sigma f(V_i)$, and

$$C_i = L_i - L_i' \quad (1 \le i \le 4), \qquad I = L_1 - \beta,$$
$$D_H = f(H), \qquad G = L_3^2 - L_2(L_2 - 2)L_3L_4 + (L_2 - 2)^2L_4^2 - 1.$$

The alternate ratios L_i' already include conjugation. The nine tests A_i, B_i have order one, C_1, I have order ten, and D_H, C_2, C_3, C_4, G have order forty. For an ordered list of tests, use the factor

$$\mathcal{T}(D_0, \dots, D_{s-1}) = 1 + \sum_{j=0}^{s-1} q_{m(D_j)}^{2^j} D_j^{2^{j+1}},$$

where $m(D_j)$ is its stated order. Thus different required orders can occur within one factor.

The detector uses the product of the factors for the following six lists:

$$(D_H, C_1, B_0), \quad (C_2, I, B_2), \quad (C_3, A_2, B_3),$$
$$(C_4, A_3, B_4), \quad (G, A_4), \quad (A_1, B_1).$$

Each of the sixteen tests occurs exactly once. The common-place valuation argument and the literal cleared trace equations are proved in [1, Proposition 2.3]; each contact prime supplies every required test order at one place. The rational logarithm approximants and the scalar point exclusions are retained.

If a nonzero integer J is divisible by the kth powers of both contact radicals, the contact-height theorem bounds $h(\lambda)$ and $h(S_0^2\lambda)$ by $(\mathcal{H} + \log 3)|J|^{19/k}$. Since $\lambda \neq 0$,

$$2h(S_0) = h(S_0^2) \le h(S_0^2\lambda) + h(\lambda).$$

Combining this with (1) gives

$$B^2 \le C|J|^{19/k} + M_2, \qquad C = M_1(\mathcal{H} + \log 3). \tag{2}$$

3.1 A quadratic presentation of the local weights

For each detector, write $F_*(t_*) = \prod_{\theta\in\Theta}(t_* - \theta)$ and put $q = R_*^{-1}\delta^{-4}$. The weights used here are $q_m = q^{2m-1}$, so $q_{10} = q^{19}$ and $q_{40} = q^{79}$. The quadratic rows are

$$q_2 = q^2, \quad q_4 = q_2^2, \quad cq_4 = q, \quad q_8 = q_4^2,$$
$$cq_{19} = q_8^2, \quad q_{38} = q_{19}^2, \quad cq_{79} = q_{38}^2.$$

At every consumed contact prime, δ is a unit and $q \neq 0$. The rows then force $c = q^{-3}$ and each indicated power uniquely. No inverse coordinate other than c is needed. If $q = 0$, the retained relations $\delta^3 q = e$ and $F_*(t_*)e = (a_*\delta)^5$ force $\delta = 0$; the contact radical is empty in that case. This covers the potentially degenerate stratum without using a rational formula outside its domain. The full local extraction proof from the literal trace equations is [1, Section 2].

At contacts these powers have exactly the required valuations: $v(q_m) = (1-2m)v_p(R_*)$. The ordered factors therefore supply the same orders $1, 10, 40$. At prepared finite primes their simultaneous smallness is established directly in [1, Section 3], by enlarging a fixed denominator before prime selection. Thus the replacement changes neither the formal logarithm comparisons nor their required depths.

3.2 Primitive normalization of the scalar contact product

Let z_j denote a coordinate by its source identifier. The actual contact product is the affine expression

$$\begin{aligned} r = & -103171881284 - 2103220824z_{248} - 21050600z_{487} \\ & - 7629700z_{488} - 45771136z_{489}. \end{aligned}$$

We use r itself as the coordinate in place of z_{489}. This is an invertible rational affine change: its coefficient of z_{489} is nonzero, and every other coordinate, including the input, is fixed. The source coordinate occurs only in its own defining row and the scalar normalizer row. After substitution and primitive normalization, the latter is exactly

$$R_s + 80989^2 r R_s - A_N r = 0.$$

Every equation remains quadratic. Reverse substitution reproduces both source rows exactly up to a nonzero rational scalar, so their rational solution sets correspond bijectively. Taking sums of squares preserves the required conjunction over $\mathbb{Q}$.

In the resulting quartic the coefficient of r^2 is $A_N^2 + 1$. Every other coefficient is either a nonzero constant or a nonzero constant multiple of A_N. Thus no coefficient vanishes for a positive parameter, and the support counts hold for every $N \geq 1$. Let C_0 be the largest absolute constant coefficient. Its exact integer value is recorded in the certificate and has 1520 binary digits. The largest absolute coefficient multiplying A_N is

$$L_0 = 2 \cdot 80989^2 = 13118436242.$$

The complete integer comparison verifies

$$80989^{94} + 1 > C_0, \qquad 80989^{47} > L_0, \qquad 80989^{90} + 1 < C_0.$$

Consequently, for both families,

$$H(P_N) = 80989^{4N+2} + 1 \qquad (N \geq 23).$$

Indeed this coefficient exceeds both C_0 and $L_0 A_N$ at $N = 23$, and continues to do so as N increases. Since $C_0 > L_0^2$, the exact height for any positive parameter is $\max(C_0, A_N^2 + 1)$. This is an algebraic height threshold; the sufficient parameter in the 342-variable defining theorem remains 27464, and the corresponding threshold for the 297-variable family remains ineffective.

4 Soundness for every sufficiently large parameter

Put $\beta = 51X^2 + 2504$ and $c = |a - 1|$. The retained scalar circuit uses the positive contact product

$$r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1)$$

and normalizer $R_s = A_N r/(1 + 80989^2 r)$, with shift $R_s + 9792$. The fixed-prime argument gives

$$c \geq 80989^{L_N}, \quad L_N = \left\lfloor \frac{187(2N-1)}{8} \right\rfloor + 1, \quad h(\beta) \leq \mathcal{H} c^{19/24}. \tag{3}$$

Its integral group identities and exceptional-point exclusions are unaffected by the local test factors.

Proof of soundness in Theorem 1.1. Set

$$\Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1.$$

The formal comparison gives divisibility of Δ by the fortieth powers of both contact radicals. If $\Delta \neq 0$, then $|\Delta| \leq 14B^4$, and (2) gives

$$B^2 \leq C14^{19/40}B^{19/10} + M_2.$$

This is impossible for sufficiently large B. Equation (3) ensures $B \geq c - 1 \to \infty$ uniformly as $N \to \infty$. Hence $\Delta = 0$ for every sufficiently large N.

The integral Lucas descent in [6] then gives $\log B > |a| - 3 \geq c - 4$. Suppose $n - \beta \neq 0$, and let J be its nonzero reduced numerator. The order-ten tests imply that both tenth powers of the contact radicals divide J, while $|J| \leq 2B\exp(h(\beta))$. For $B^2 \geq 2M_2$, equation (2) therefore implies

$$\log B \leq 10\log(2C) + 19\log 2 + 19h(\beta) \leq 10\log(2C) + 19\log 2 + 19\mathcal{H}c^{19/24}.$$

The right side is $o(c)$, contradicting the Lucas lower bound for sufficiently large c. Thus $n = \beta \in \mathbb{Z}$. Writing $X = u/v$ in lowest terms, $v > 0$, gives $v^2 \mid 51$ and hence $v = 1$.

All thresholds depend only on the fixed constants $C, M_2, \mathcal{H}$. Since L_N increases, one integer $N_0 \geq 27464$ works for every $N \geq N_0$. The dependence on M_2 is precisely the ineffective part of the assertion. □

5 Completeness and exact verification

Fix an integer input and the scalar parameter. The retained scalar and Lucas construction supplies the integer labels and nonzero root points. Choose compact smaller real and finite-place neighborhoods avoiding the poles of the retained rational tests. The tests then have fixed valuation lower bounds at every prime in the finite preparation set S. All scales and neighborhoods are fixed before the contact primes.

Choose a positive integer D that clears the contact coefficients $D^2\theta/a_*$ and has sufficiently large valuations at the primes in S. In the simultaneous prime pattern put

$$\lambda = \frac{u}{D^2w^2}.$$

The zero contact makes u a selected prime, outside S. Therefore $v_p(\lambda) \leq -2v_p(D)$. Making this upper bound sufficiently negative gives $v_p(t_*) < \min(0, v_p(\theta) : \theta \neq 0)$ and hence

$$v_p(F_*(t_*)) = 5v_p(t_*), \quad v_p(e) = 5v_p(a_*), \quad v_p(q) = 2v_p(a_*) - 3v_p(\lambda).$$

The latter is uniformly arbitrarily large. Thus all new weighted test terms are arbitrarily small at the prepared finite places. Enlarging D is a fixed dilation of the prime pattern and is allowed by the retained selection theorem; see [2].

At each selected contact prime, the designated primary place gives all test orders. The distinct-valuation argument at the opposite place makes $\mathrm{Tr}(W^{-1})$ a unit congruent to 1 or 2 modulo $p > 2$. Consequently the pole input $R_*\,\mathrm{Tr}(W^{-1})^{-2}$ has valuation one. The exact product formula

$$R_* = \frac{\prod_{\theta\in\Theta}(u - (\theta/a_*)D^2w^2)}{(D^2w^2)^5\delta^6}$$

shows that its valuation is even at every other prime. This includes all primes dividing δ; no unit assertion about δ away from contacts is required. Its reduced denominator is therefore a

square. The selected-prime congruences give the required local squareclasses at $2, 3, 17$, and the fixed choice of D gives their required valuations.

Uniformly on the fixed compact real neighborhoods,

$$R_* \sim a_*^{-6}\lambda^{-1}, \qquad q \sim a_*^2\lambda^{-3}, \qquad \mathrm{Tr}(W^{-1}) \longrightarrow 2.$$

All factors tend to 1, and the pole input tends to 0^+. It therefore lies in $(0, 17)$ for sufficiently large λ. The simultaneous selection supplies such parameters after all fixed data have been chosen. The pole-realization premise gives fresh witnesses, and the rational weight identities fill every new quadratic row. Equal scales may reuse witnesses. This is the full simultaneous completeness argument of [1, Proposition 3.2]; it imposes no new lower bound on the scalar parameter N.

The exact reproduction programs are

```
python build_quartic_shared_weights.py
python check_quartic_shared_weights.py
```

The checker imports no producer. It checks every changed and retained row, all coordinate incidences, and every coefficient of the quartic by an independent unordered-pair expansion. It verifies degree four, content one and the coefficient-height bound. These finite checks do not prove the retained global arithmetic inputs.

The exact table contains 271 rows, 297 total coordinates, 4269 terms before parameter powers are collected, and 4268 ordinary monomials for every $N \geq 1$. The weight presentation uses one coordinate c and one coordinate q_8 in each detector, together with the five displayed power outputs. The coefficient family is fully specified; the sufficient defining threshold remains ineffective because of the aggregate trace height.

References

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[7] *Aggregate trace heights of elliptic multiples*, 8 November 2026, proof of the linear height comparison from Von Buhren's family height theorem. PDF; source: `aggregate_trace_height_20261108_v9.tex`.

[8] E. Glazer, *An explicit quartic defining the integers in the rationals*, Version 13, manuscript and coefficient data, commit `5bbd74f6a479776a332cbe036809d5dddae63720`, https://github.com/elliotglazer/diophantine-z-q.